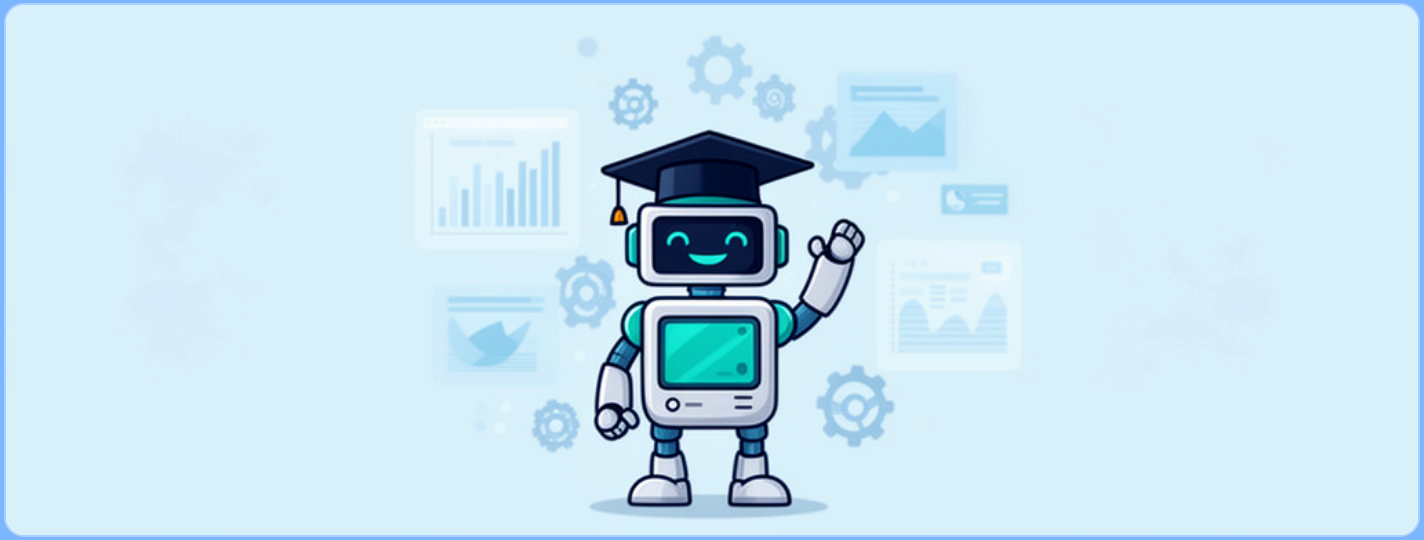
How the Tree Picks Its Questions

Day Three — You Become the Question-Picker

Block 1 · Session 3

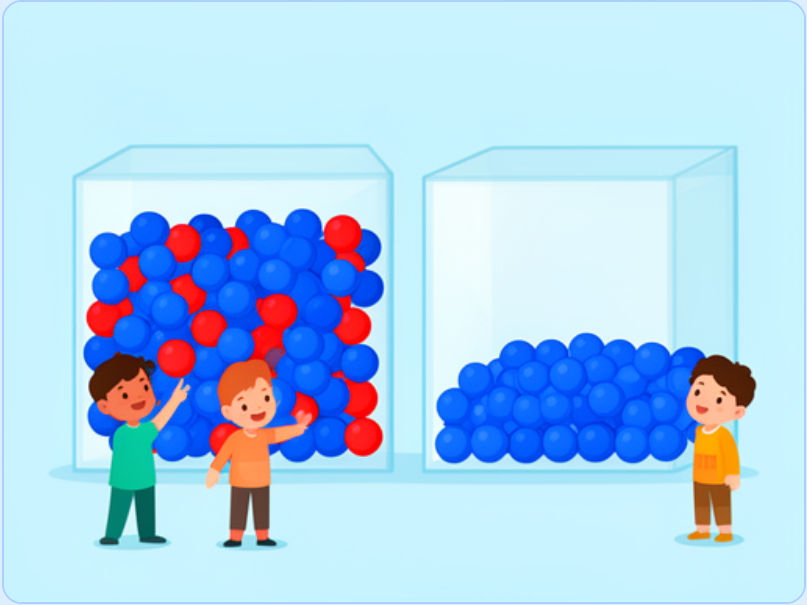
.Today you learn which question is better — and why



Two Boxes, One Question

.Goal: every box becomes **one color**

?Which box needs help? Which one is done



?Which Question Sorts Best

.A new box: 2 blue + 2 red

"?Vote: "is it red?" or "is it big

.Keep your votes — the machine plays this exact game

The Machine Scores Its Questions

?Last session the machine asked `weight >= 14`. **Why 14**

.Watch: it tries every question and gives each a score

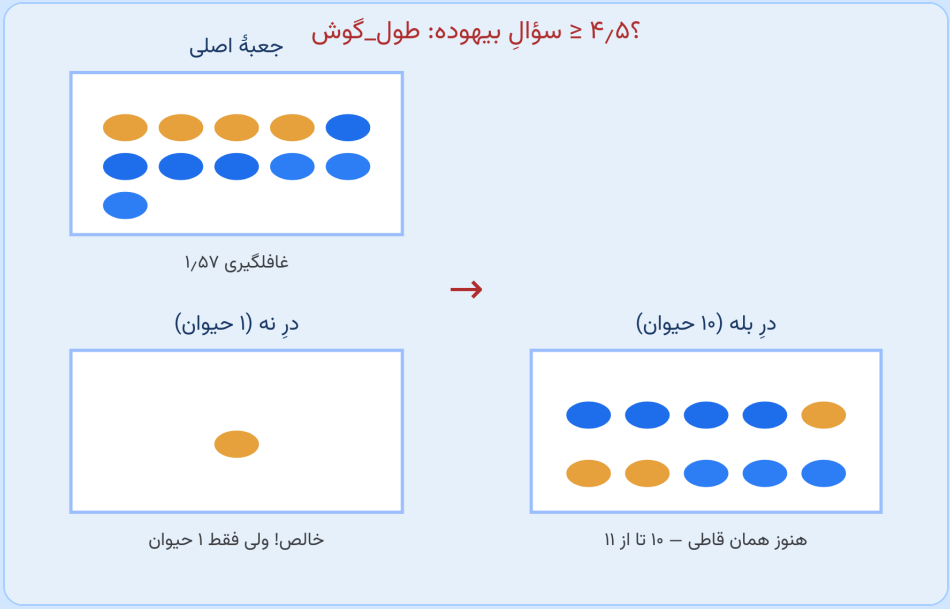


Sorted-ness (Purity)

.A good question leaves each side as **one thing**

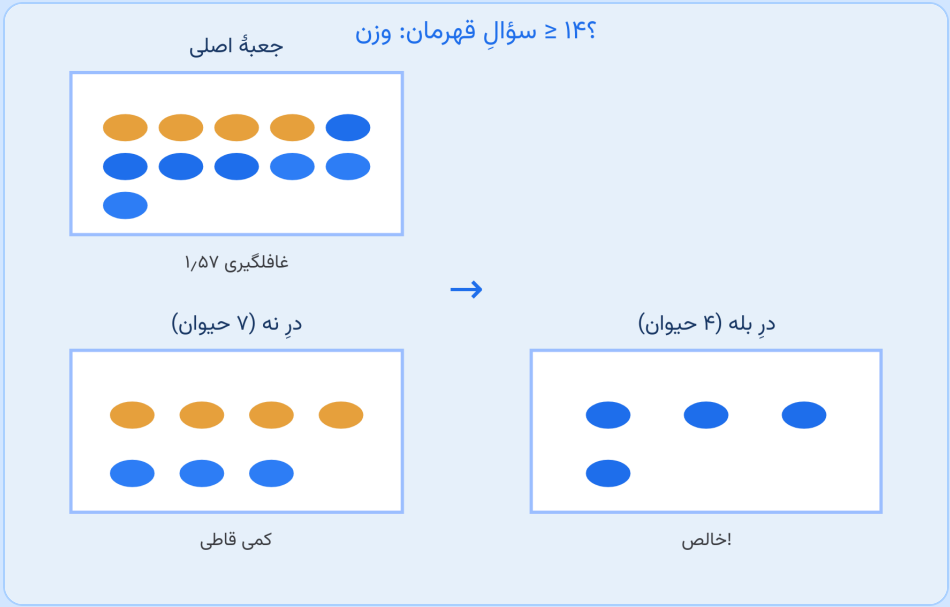
.A useless question cuts the box but keeps the mess

The Useless Question



.The no-door IS pure — but holds only 1 animal! The other 10 stay just as mixed

The Champion Question



!The yes-door: only big dogs. **Pure**

?Sorted or Just Cut

no: 9 mixed animals · yes: 2 big dogs ← `weight >= 30`

?Did it sort — or only cut

The Grab Bags

.Three bags, one blind draw each

?Before every draw: say your guess. Then: how surprised were you



Surprise (Entropy)

:The box's messiness number

- pure box ← 0
- more mixed ← bigger number

.It comes from counting colors. No formula needed



?The 11-Animal Box: How Much

.cats, 4 big dogs, 3 small dogs 4

.Guess the surprise first, then check

Game: You Give the Scores

?On the board: each question → two doors → who stands on each side

.Teams of two: compute "mess before minus mess after" for every question

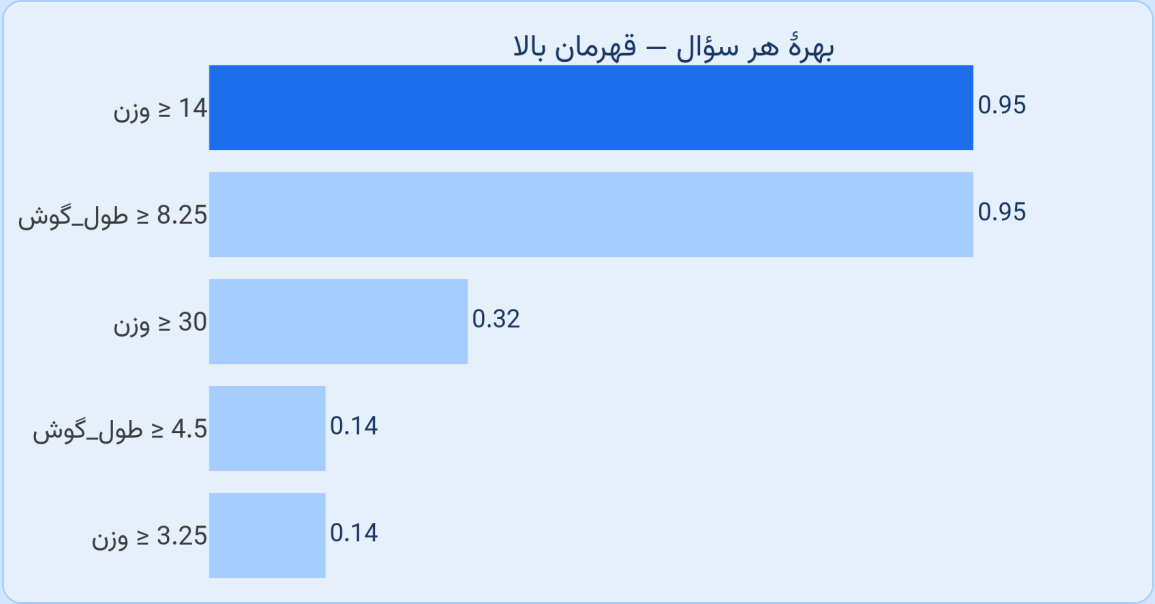
!Shout the champion

A Question's Score: Information Gain

gain = surprise before − surprise after

.The champion question: the biggest gain

The Leaderboard



. `ears >= 8.25` The champions TIE: `weight >= 14` and

Round Two: The Light Box

animals: 4 cats + 3 small dogs. Surprise: 0.99 7

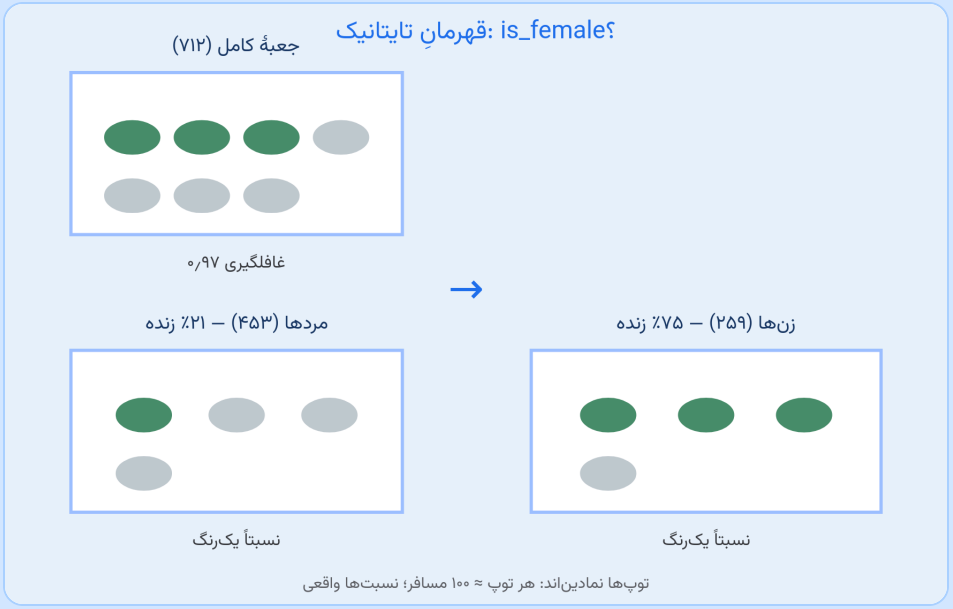
?Who is THIS box's champion

The Machine Plays It on Titanic

.passengers. A mixed box: 288 survived, 424 did not 712

.Session 1's data — this time with a score

Titanic's Champion



.gain 0.21, nearly three times the runner-up (0.08) — `is_female`

The Whole Secret

~/ml-course

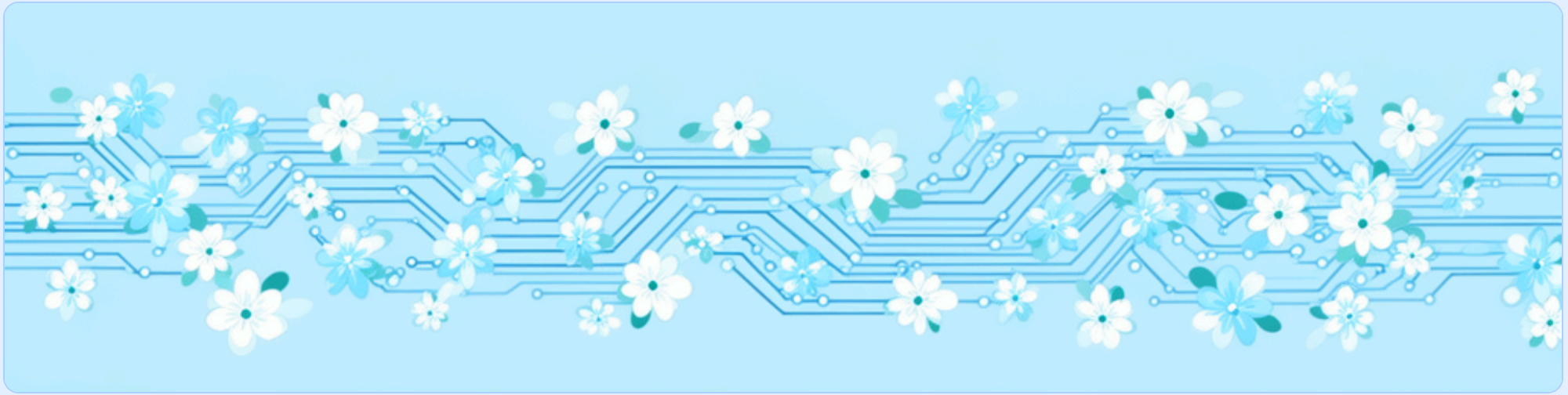
```
for every possible question:
    score = surprise_before - surprise_after
champion = the biggest score
```

.There is no line four

Choosing Features Is OUR Job

.The machine only shops among the table's columns

.Does it hop?" is never asked — it isn't a column"



What You Learned Today

- **Purity** = one-color boxes
- **Entropy** = the surprise number
- **Gain** = before − after; champion = biggest gain
- Titanic's champion: `is_female` — Session 1's vote

Tonight's Delivery + At-Home (3)

B1 The surprise of three bags 2. **B2** The light box champion 3. **B3** Titanic's champion + the **.1** split plot

instant score, anytime ← `python judge/run.py s03_notebook.ipynb`

My Session 1 vote got its number today." — **your third Ship**"



Next Session

...The machine scored its questions on passengers it has SEEN

?What if it only memorized them

.Next time: the exam cheat test

